

WAWA, ON, Canada

WMO: 717380

Lat: 47.9730N

Lon: 84.7824W

Elev: 287

StdP: 97.92

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.8	-34.8	0.1	-30.9	-31.5	0.2	-27.5	10.2	-5.5	9.2	-5.3	1.0	20	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	25.3	18.7	23.6	17.9	22.3	17.1	20.3	23.2	19.3	22.1	18.4	21.0	3.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.6	21.8	18.3	13.6	20.7	17.4	12.8	19.9	60.0	23.2	56.4	22.3	53.3	21.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.3	6.4	-35.6	28.6	2.9	1.4	-37.7	29.6	-39.3	30.4	-41.0	31.1	-43.1	32.1		
			-35.9	22.6	2.8	0.9	-37.9	23.2	-39.5	23.7	-41.1	24.2	-43.1	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-13.2	-11.6	-6.1	1.1	8.2	13.3	15.4	15.7	12.2	5.7	-1.3	-8.4
	DBStd	11.53	7.84	7.00	6.92	4.71	4.37	3.67	3.06	2.90	4.16	4.46	5.80	7.13
	HDD10.0	3277	720	605	498	268	88	11	2	1	24	149	341	570
	HDD18.3	5749	978	838	756	517	314	155	100	92	187	392	591	828
	CDD10.0	595	0	0	0	1	33	111	168	176	91	16	1	0
	CDD18.3	25	0	0	0	0	1	4	8	9	3	0	0	0
	CDH23.3	167	0	0	0	0	19	42	47	43	14	1	0	0
	CDH26.7	13	0	0	0	0	2	4	3	4	1	0	0	0
Wind	WSAvg	2.6	2.7	2.8	2.7	2.8	2.5	2.2	2.1	2.3	2.6	2.7	2.8	2.7
Precipitation	PrecAvg	1020	80	57	55	71	83	80	87	85	122	116	88	95
	PrecMax	1402	156	155	104	150	211	159	163	191	261	214	157	204
	PrecMin	483	41	19	8	16	26	18	12	19	38	39	27	41
	PrecStd	221	26	27	24	38	40	39	37	40	53	50	30	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	6.5	13.4	18.7	25.9	27.0	27.0	26.8	25.3	20.8	13.1	5.9
		MCWB	2.4	4.6	8.0	10.5	17.9	18.3	19.7	20.2	18.9	16.0	10.4	4.9
	2%	DB	1.8	2.7	8.5	14.0	22.0	24.4	24.7	24.3	22.7	18.0	10.5	4.0
		MCWB	1.0	1.4	5.1	7.2	14.4	17.4	18.8	19.1	18.1	14.2	8.5	2.9
	5%	DB	0.5	0.7	5.5	11.0	19.2	22.6	23.0	22.7	20.9	15.6	8.4	2.6
		MCWB	-0.2	-0.3	2.8	5.9	12.7	16.2	17.8	18.1	17.5	13.0	6.7	1.7
	10%	DB	-1.2	-0.9	3.1	8.6	16.6	20.8	21.5	21.3	19.2	13.1	6.5	1.2
		MCWB	-2.0	-1.9	1.0	4.5	11.3	15.1	17.1	17.3	16.3	11.0	5.1	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	4.9	9.2	11.2	19.0	21.1	21.4	21.8	20.5	17.0	11.2	5.2
		MCDB	3.1	5.8	11.1	15.8	24.1	24.3	24.6	24.6	22.7	19.2	12.8	5.5
	2%	WB	1.1	1.6	5.5	8.6	16.3	19.2	20.0	20.2	19.2	14.7	8.9	3.1
		MCDB	1.7	2.5	7.8	12.4	20.4	22.4	23.0	22.9	21.7	17.3	10.3	3.7
	5%	WB	-0.1	-0.1	3.1	6.7	14.0	17.7	18.9	19.1	18.0	13.1	6.8	1.7
		MCDB	0.5	0.6	4.8	10.3	17.5	20.7	21.7	21.7	20.1	15.3	8.1	2.5
	10%	WB	-2.0	-1.9	1.2	4.7	12.0	16.2	17.9	18.2	16.7	11.1	5.2	0.5
		MCDB	-1.3	-0.9	2.7	8.1	15.3	19.3	20.6	20.4	18.7	13.2	6.4	1.2
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.5	11.7	10.7	12.0	11.8	10.2	9.7	9.4	8.0	7.1	8.7
		MCDBR	7.3	9.9	11.5	14.9	16.5	14.7	12.4	11.6	11.0	11.4	8.8	7.6
		MCWBR	7.1	8.9	8.1	9.3	9.4	8.3	7.3	7.0	6.9	8.0	7.3	7.1
	5% WB	MCDBR	7.3	9.4	10.2	13.6	13.6	12.1	10.4	9.7	9.5	10.1	8.2	7.3
		MCWBR	7.4	8.7	7.9	9.3	9.0	8.1	7.1	6.5	6.8	7.8	7.3	7.0
Clear-Sky Solar Irradiance	taub	0.273	0.271	0.308	0.335	0.360	0.377	0.387	0.367	0.365	0.325	0.290	0.297	
	taud	2.309	2.319	2.320	2.388	2.404	2.380	2.369	2.428	2.433	2.529	2.522	2.237	
	Ebn at Noon	804	897	912	918	902	885	869	874	841	829	794	705	
	Edh at Noon	72	89	106	110	113	116	116	105	95	74	60	69	
All-Sky Solar Radiation	RadAvg	1.11	2.05	3.46	4.68	5.44	5.84	5.81	4.89	3.55	2.10	1.05	0.79	
	RadStd	0.16	0.24	0.33	0.46	0.46	0.39	0.41	0.36	0.25	0.27	0.13	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page